MILITARY SPECIFICATION SHEET

COUPLERS, DIRECTIONAL (WAVEGUIDE-CROSSGUIDE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the couplers described herein shall consist of this document and the latest issue of Specification MIL-C-15370.

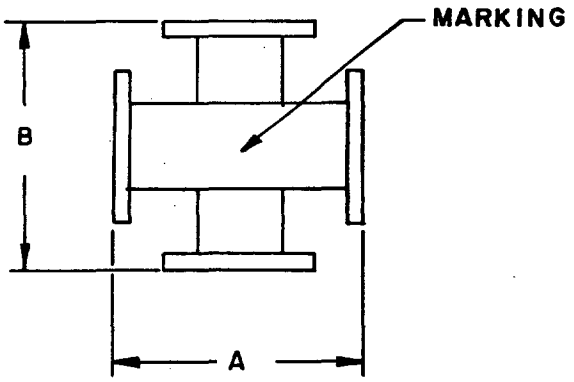


FIGURE 1. Dimensions and configuration for part numbers M15370/5-001 and -002.

FSC 5985

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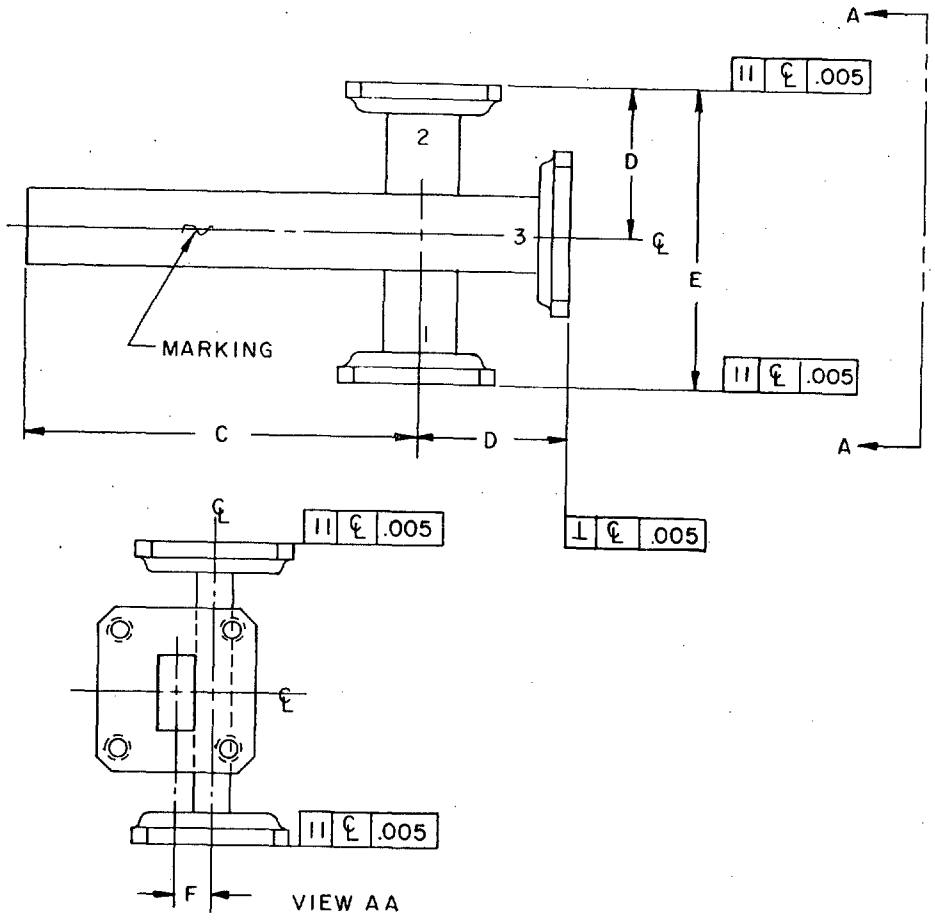


FIGURE 2. Dimensions and configuration for part numbers M15370/5-003 thru -006.

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REQUIREMENTS:

Design, construction, and physical dimensions: See figures 1 and 2, and table I.

Equivalent waveguide: See table II.

Flange: See table II.

Temperature range: -65°C to 95°C.

Coupling: See table II.

Effective directivity: See table II.

VSWR: See table II.

Frequency range: See table II.

Power rating: See table II.

Marking: See figures 1 and 2.

Part number: M15370/5-(and a dash number from table I).

TABLE I. Physical characteristics.

Dash number	Dimensions 1/ 2/ 3/							
	A		B		C	D	E	F
	Min	Max	Min	Max	Max	±.02	±.02	±.02
001	5.938 (150.83)	6.063 (154.00)	5.938 (150.83)	6.063 (154.00)				
002	2.938 (74.63)	3.063 (77.80)	2.938 (74.63)	3.063 (77.80)				
003	---	---	---	---	3.50 (88.90)	1.25 (31.75)	2.50 (63.50)	.39 (9.91)
004	---	---	---	---	3.50 (88.90)	1.25 (31.75)	2.50 (63.50)	.39 (9.91)
005	---	---	---	---	3.50 (88.90)	1.25 (31.75)	2.50 (63.50)	.39 (9.91)
006	---	---	---	---	3.50 (88.90)	1.25 (31.75)	2.50 (63.50)	.39 (9.91)

1/ Dimensions are in inches.
2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.

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TABLE II. Electrical characteristics.

Dash number	Frequency range	Coupling factor	Mean coupling 1/	Coupling variation 1/	Effective directivity (min) 1/	VSWR (max)	Power dissipation of internal load (avg)	Equivalent waveguide	Flange
	GHz	dB	dB	dB	dB		watts		
001	3.95- 5.85	20	±0.5	±1.7	20	1.05	2	M85/1-055 (RG-49/U)	M3922/57-002 (UG-149A/U)
002	8.2 -12.4	20	±0.5	±1.7	20	1.05	1	M85/1-079 (RG-52/U)	M3922/53-001 (UG-39/U)
003	12.4 -18.0	20	±0.5	3 max 2/	20	1.15	1	M85/1-090 (RG-349/U)	M3922/53-006 Port 1 (UG-1665/U) M3922/59-002 Ports 2 and 3 (UG-1666/U)
004	12.4 -18.0	30	±0.5	3 max 2/	20	1.10	1	M85/1-090 (RG-349/U)	M3922/53-006 Port 1 (UG-1665/U) M3922/59-002 Ports 2 and 3 (UG-1666/U)
005	12.4 -18.0	20	±0.5	3 max 2/	20	1.15	1	M85/1-090 (RG-349/U)	M3922/59-002 All ports (UG-1666/U)
006	12.4 -18.0	20	±0.5	3 max 2/	20	1.15	1	M85/1-090 (RG-349/U)	M3922/53-012 Ports 1 and 2 UG-1665/U Port 3 3/

1/ Measured by swept frequency techniques.
2/ Total change across the frequency band.
3/ Except the lip thickness is .25" and tapped holes .138-32UNC-2B

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TABLE III. Cross reference of part number/AN nomenclature.

Dash number	AN nomenclature
001	CU-1506/U
002	CU-988/U

Note: Directional couplers covered by this specification sheet are substitutable for the commercial types listed in table IV.

TABLE IV. Substitution data.

Part number M15370/4-	FSN 5985-	Manufacturer's code	Commercial type
001	982-3911	28480	G750D
001	982-3911	87864	TR78217
002	261-5823	28480	X750D
002	889-4217	28480	X750D
002	889-4217	98732	12J44
003		05869	723372-1
003		00443	723372-1
003		17250	723372-1
004		98732	723372-2
004		00443	723372-2
004		17250	723372-2
005		98732	723372-3
005		00443	723372-3
005		00443	723372-3
006		98732	723372-4
006		98732	723372-4
006		98732	723372-4

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - SG
Navy -
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - WC
Navy - AS, CG, MC, SH, OS
Air Force - None

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0786)

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